Geometric Observables for Financial Regime Detection: A Spectral Metric Learning Approach

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Abstract

We extract geometric observables from a spectral metric learning framework—Quantum Cognition Machine Learning (QCML; the name is historical—the mathematics is spectral geometry, not quantum physics)—and evaluate their ability to separate crisis from non-crisis periods in equity markets. QCML encodes data as quasi-coherent states in a Hilbert space via an error Hamiltonian; the induced Fubini–Study metric and Berry curvature provide three scalar observables: the Berry curvature increment, the quantum Fisher information (QFI) determinant, and a multi-lag fidelity measure.

In a *crisis-window separability* study of 12 historical crises (2007–2024) using per-crisis causal preprocessing (scaler, PCA, and operators fitted only on pre-crisis data), these observables—whose score construction is unsupervised—achieve median Cohen's $d = 0.56$ (Multi-Lag Fidelity), $d = 0.42$ (Berry), and $d = 0.40$ (QFI Determinant). Multi-Lag Fidelity matches the supervised Random Forest baseline ($d = 0.96$) on mean Friedman rank (both 4.42), despite requiring no labeled data. The best geometric observable outperforms RF on 5 of 12 crises—precisely those where RF labels are sparse or absent (e.g., 2023 SVB: MLF $d = 0.94$ vs. RF $d = 0.12$). CUSUM ($d = 0.73$) remains the strongest classical baseline.

A walk-forward causal benchmark with adaptive threshold calibration (rolling quantile + score velocity) achieves 86% detection rate with 9-day median delay for Multi-Lag Fidelity, transforming the offline separability into a viable early-warning signal. Hyperparameters are selected via Optuna TPE with a consistency penalty, and every Cohen's d includes a bootstrap 95% CI ($n = 5{,}000$).

Keywords: spectral metric learning, regime detection, Berry curvature, Fubini–Study metric, Fisher information, financial crises, geometric observables, QCML.

1 Introduction

Detecting regime transitions—shifts between qualitatively different market states—is a central problem in quantitative finance. Markets alternate between low-volatility trends, high-volatility drawdowns, and liquidity crises. Classical approaches include Hidden Markov Models [9], Bayesian Online Changepoint Detection [10], and supervised classifiers. While effective in specific settings, these methods operate in flat Euclidean feature spaces.

Quantum Cognition Machine Learning (QCML) [1, 3] encodes data points as unit vectors in a finite-dimensional Hilbert space and represents features as Hermitian operators. The map $x \mapsto |\psi(x)\rangle$ from feature space to projective Hilbert space $\mathbb{CP}^{n-1}$ endows the data manifold with a

Fubini–Study metric, Berry curvature, spectral gap, and quantum Fisher information. Candelori et al. [1] showed that this geometry robustly captures intrinsic dimension; Abanov et al. [2] developed the mathematical theory. Financial applications include forecasting [4], similarity learning [5, 6], and biomedical prediction [7].

Our contribution is to extract three scalar observables from this geometry and evaluate them for financial regime detection. The evaluation is framed as *crisis-window separability*: given a score time series, we ask whether the distribution of scores during a known crisis window is statistically separated from the non-crisis distribution. This is an *offline event study*—it measures sensitivity to regime transitions, not real-time predictive power. A supplementary walk-forward benchmark (Section 5.4) tests causal detection with strictly past-fitted preprocessing.

Clarification on "unsupervised." Score construction is unsupervised: the QCML observables require no labeled crisis data to compute z-scores (Algorithm 1). Hyperparameters (Hilbert dimension, PCA components, operator method, rolling window) are selected via Optuna TPE with a consistency penalty (Berry: $n = 6$, $p = 8$, $w = 15$; QFI: $n = 8$, $p = 15$, $w = 20$; Multi-Lag: $n = 4$, $p = 8$, $w = 20$); operator method is selected by ablation on historical data (Section 5.6). We use "unsupervised score construction with HPO-optimized hyperparameters" throughout.

Contributions.

1. Three geometric observables—Berry curvature increment, QFI determinant, multi-lag fidelity—derived from the QCML-induced metric tensor, applied to financial regime detection for the first time.

2. A reproducible evaluation protocol with HPO-optimized hyperparameters (Table 3), per-crisis causal preprocessing, bootstrap confidence intervals on all effect sizes, and expanding-window walk-forward validation.

3. A walk-forward detection benchmark and a two-way ANOVA testing for complementarity between geometric and classical methods across novel vs. conventional crises.

We emphasize that the contribution is *not* that geometric observables uniformly dominate all existing methods—CUSUM (median $d = 0.73$) and RF ($d = 0.96$) outperform geometric methods on aggregate median [20, 22]. Rather, the contribution is threefold: (1) a new class of manifold-intrinsic observables for regime detection, grounded in the differential geometry of projective Hilbert space, that achieve competitive Friedman rank (MLF ties RF at 4.42) without any labeled data; (2) evidence that these observables exhibit complementary per-crisis strengths, winning 5 of 12 crises where RF labels are sparse or absent; and (3) a rigorous causal evaluation protocol that enables honest assessment of where geometric methods add value and where classical methods remain stronger.

1.1 Related Work

Several non-Euclidean frameworks have been applied to financial data. We briefly survey the most relevant and position our contribution.

Information geometry. Amari and Nagaoka [15] introduced the Fisher–Rao metric on statistical manifolds; Brody and Hughston [28] applied it to interest rate term structures, and Taylor [29] used Fisher distance for clustering return distributions. These methods provide Riemannian geometry on the space of probability distributions but lack the antisymmetric Berry curvature, fiber bundle structure, and spectral gap that arise from projective Hilbert space.

Optimal transport. Horvath et al. [30] cluster market regimes via Wasserstein distance. Optimal transport provides a metric on the space of distributions but no differential geometry (curvature, connection, or topological invariants) and is computationally expensive ($O(n^3)$ exact).

TDA and persistent homology. Gidea and Katz [31] detect crash precursors via L^p-norms of persistence landscapes. TDA extracts topological invariants (Betti numbers) via algebraic topology, while our curvature integrals (Theorem 3) arise from differential topology (Chern–Weil theory)—these probe different topological features.

Network curvature. Sandhu et al. [32] use Ollivier–Ricci curvature on stock correlation networks as a systemic risk indicator—the closest curvature-based regime detector to our work. Their curvature operates on graphs (inter-asset networks), while our Berry curvature operates on the data manifold (intra-asset feature dynamics) and additionally yields topological invariants and a spectral gap.

Random matrix theory. Laloux et al. [33] and Plerou et al. [34] use the Marchenko–Pastur distribution to separate signal from noise in correlation matrices. Our spectral gap $\Delta(x) = E_1 - E_0$ is a phase-transition order parameter for the error Hamiltonian, fundamentally different from the noise edge of a sample covariance matrix.

Rough path signatures and geometric deep learning. Issa and Horvath [35] detect regimes via signature-based MMD tests; signatures provide universal path features but lack geometric interpretability. Bronstein et al. [36] systematize gauge-equivariant neural architectures but impose geometric priors rather than discovering geometry from data.

Table 1 summarizes these distinctions. Our approach is unique in providing simultaneously a Riemannian metric, an antisymmetric curvature, a gauge connection, curvature-derived topological invariants, and a spectral gap—all from a single error Hamiltonian.

Table 1: Non-Euclidean approaches to financial regime detection. Columns indicate which geometric tools each framework provides.

Framework	Metric	Curvature	Gauge	Topol. Invt.	Spect. Gap
Info. Geometry	Fisher–Rao	Sectional	α-conn.	—	—
Optimal Transport	Wasserstein	—	—	—	—
TDA	—	—	—	Betti	—
Network Ricci	—	Ollivier	—	—	—
Random Matrix Thy.	—	—	—	—	MP edge
Rough Path Sig.	Sig. kernel	—	—	—	—
QCML (ours)	**Fubini–Study**	**Berry**	**U(1)**	**Chern**	$\boldsymbol{\Delta(x)}$

2 QCML Framework

We work in a finite-dimensional Hilbert space $\mathcal{H} \cong \mathbb{C}^n$. A *state* is a unit vector $|\psi\rangle \in \mathcal{H}$, and an *observable* is a Hermitian operator $A = A^\dagger$ [1, 2].

Given $x = (x_1, \ldots, x_p) \in \mathbb{R}^p$ and Hermitian operators $\{A_k\}_{k=1}^p$, the QCML *error Hamiltonian* [1, 3] is:

$$H(x) = \frac{1}{2} \sum_{k=1}^{p} (A_k - x_k I)^2 , \tag{1}$$

where I is the $n \times n$ identity. The *quasi-coherent state* at x is the ground state:

$$|\psi(x)\rangle = \underset{|\phi\rangle,\, \|\phi\|=1}{\arg\min} \langle \phi | H(x) | \phi \rangle . \tag{2}$$

The pullback of the Fubini–Study metric defines the *quantum metric tensor* [14]:

$$g_{ab}(x) = \mathrm{Re} \langle \partial_a \psi | \partial_b \psi \rangle - \langle \partial_a \psi | \psi \rangle \langle \psi | \partial_b \psi \rangle , \tag{3}$$

where indices $a, b \in \{0, \ldots, p-1\}$ run over PCA components, and the *Berry curvature* [2, 8]:

$$F_{ab}(x) = -2\, \mathrm{Im} \langle \partial_a \psi | \partial_b \psi \rangle . \tag{4}$$

Theorem 1 (Smoothness [1])**.** *If the spectral gap $\Delta(x) = E_1(x) - E_0(x) > 0$ on an open set $U \subset \mathbb{R}^p$, then $x \mapsto |\psi(x)\rangle$, $g_{ab}(x)$, and $F_{ab}(x)$ are C^∞ on U.*

Proof sketch. By the implicit function theorem applied to the eigenvalue equation $H(x)|\psi\rangle = E_0|\psi\rangle$, $\langle \psi | \psi \rangle = 1$; the non-degenerate spectral gap ensures invertibility of the relevant Jacobian. See Appendix A for the full argument. □

Operator construction. We use *PCA-inspired* operators: Pauli-basis matrices scaled by PCA eigenvalues, $A_k = \sqrt{\lambda_k}\, \sigma_k$. An ablation comparing random, PCA-inspired, and learned-scaling operators appears in Section 5.6. Full QCML gradient-descent training [3, 4] is left for future work.

Proposition 2 (Fisher information interpretation)**.** *For the pure states $|\psi(x)\rangle$ arising as ground states of $H(x)$, the quantum metric tensor* (3) *satisfies $4\, g_{ab}(x) = [F_Q]_{ab}(x)$, where F_Q is the quantum Fisher information matrix [26]. Consequently:*

(i) The QFI determinant observable (Eq. 6) is proportional to the volume element of the Fisher information metric on the statistical manifold: $\det^+(g) = 4^{-r} \det^+(F_Q|_r)$*, where* $r = \mathrm{rank}(g)$*.*

(ii) The quantum Cramér–Rao bound $\mathrm{Var}(\hat{x}_a) \geq \frac{1}{4}[g^{-1}]_{aa}$ *implies that the quantum metric controls the ultimate precision for estimating data coordinates from state measurements [27].*

(iii) Near regime boundaries where $|\psi(x)\rangle$ changes rapidly, $\det^+(g)$ *increases, reflecting increased distinguishability of neighboring data distributions: $D_{\mathrm{KL}}(P_x \| P_{x+dx}) \approx 2\, dx^T g\, dx$.*

Proof. Part (i) follows from the Braunstein–Caves theorem [26] and Proposition 5. Part (ii) is the quantum Cramér–Rao bound [27]. Part (iii) follows from the local expansion of KL divergence in terms of the Fisher metric [15]. □

3 Geometric Observables

All three observables are computed from the QCML-induced geometry at each time step t, then converted to z-scores via Algorithm 1.

3.1 Berry Curvature Increment

The rate of change of Berry curvature between consecutive time steps:

$$\dot{\gamma}(t) = |F_{01}(x_t) - F_{01}(x_{t-1})| \,, \tag{5}$$

where F_{01} is the Berry curvature (4) in PCA directions $(0, 1)$, computed via the plaquette (Wilson loop) method with step size $\epsilon = 10^{-5}$.

Hyperparameters. Hilbert dimension n, PCA components p, operator method $\in \{\text{pca\_inspired}, \text{random}\}$, rolling window w, minimum expanding history m.

3.2 QFI Determinant

The pseudo-determinant of the quantum metric tensor:

$$\overset{+}{\det}(g(x_t)) = \prod_{\lambda_i > \epsilon_{\text{tol}}} \lambda_i \,, \tag{6}$$

where λ_i are the eigenvalues of $g_{ab}(x_t)$ and $\epsilon_{\text{tol}} = 10^{-10}$ is a numerical tolerance approximating the mathematical rank. Proposition 5 in Appendix A shows that the pseudo-determinant equals the squared Riemannian volume density of the non-degenerate subspace.

Hyperparameters. Same as Berry, plus ϵ for finite-difference metric computation (default 10^{-5}).

3.3 Multi-Lag Fidelity

Weighted average infidelity across time lags:

$$\bar{\mathcal{I}}(t) = \sum_j w_j \big(1 - |\langle \psi(x_t) | \psi(x_{t-\delta_j}) \rangle|^2\big) \,, \tag{7}$$

with $\delta \in \{1, 3, 5, 10\}$ and weights $w = [0.4, 0.3, 0.2, 0.1]$. Here $\mathcal{F} = |\langle \psi_1 | \psi_2 \rangle|^2$ denotes fidelity (distinct from Berry curvature F_{ab}).

Hyperparameters. Same as Berry, plus lag set and weights.

3.4 Z-Score Thresholding (Algorithm 1)

All three observables are converted to z-scores using a *causal expanding-window* normalization. This is the "adaptive thresholding" referenced in Algorithm 3.
Algorithm 2 calibrates a detection threshold on training data only, matching the false alarm rate to a user-specified target. We use $\alpha = 1$ alarm/year as the default. This avoids the scale mismatch of a fixed $z > 2.0$ threshold across methods whose score distributions differ.
Defaults: $w = 20$, $m = 60$; tuned via the HPO protocol in Section 4.2. No future data enters the z-score computation.

Algorithm 1 Causal Expanding-Window Z-Score

Require: Raw observable series $r(1), \ldots, r(T)$; rolling window w; minimum history m
Ensure: Z-score series $z(m), \ldots, z(T)$
Smooth: $s(t) = \frac{1}{w} \sum_{i=0}^{w-1} r(t-i)$ for $t \geq w$
for $t = m, \ldots, T$ **do**
  $\mu(t) = \text{mean}(s(1), \ldots, s(t-1))$ ▷ expanding mean of past only
  $\sigma(t) = \text{std}(s(1), \ldots, s(t-1))$ ▷ expanding std of past only
  $z(t) = \big(s(t) - \mu(t)\big)/\sigma(t)$
end for

Algorithm 2 FAR-Calibrated Threshold

Require: Training z-scores $z(1), \ldots, z(T_{\text{train}})$; target FAR α (alarms/year); known crisis windows
Ensure: Threshold τ
Remove crisis-window indices from z to obtain z_{normal}
Binary search: find smallest τ such that $\text{FAR}(z_{\text{normal}} > \tau) \leq \alpha$
return τ

4 Experimental Protocol

4.1 Data and Crisis Definitions

Daily OHLCV data for SPY and DIA (2005–2024) from the Polygon.io API. Features: log returns, 5/20-day rolling volatility, 5/20-day momentum, cross-correlation, and cross-dispersion (13 raw features), enriched to 52 features via rolling statistics (mean, std, skew, kurtosis over a 20-day lookback), then reduced to $p = 15$ PCA components and L^2-normalized.

We evaluate on 12 historical crises (Table 2). Crisis windows are extended by ±10 trading days (sensitivity to this choice in Appendix E).

Table 2: Crisis periods. Categories defined *a priori* for the interaction test (Section 5.5).

Crisis	Period	Category	Rationale
2007 Quant Meltdown	2007-08 to 2007-09	Conventional	Resembles LTCM
2008 GFC	2008-09 to 2009-03	Conventional	Credit crisis
2010 Flash Crash	2010-05 to 2010-06	Conventional	Resembles 1987
2011 Euro Crisis	2011-07 to 2011-10	Conventional	Sovereign debt
2015 China Crash	2015-07 to 2015-09	Conventional	EM contagion
2020 COVID	2020-02 to 2020-04	Conventional	Exogenous shock
2018 Volmageddon	2018-01 to 2018-04	Novel	Short-vol blowup
2018 Q4 Selloff	2018-10 to 2018-12	Novel	Algo-driven
2019 Repo Crisis	2019-09 to 2019-10	Novel	Plumbing crisis
2022 Rate Hikes	2022-01 to 2022-10	Novel	Fastest in 40yr
2023 SVB	2023-03 to 2023-04	Novel	Social-media bank run
2024 Carry Unwind	2024-07 to 2024-08	Novel	Yen carry

4.2 Methods and HPO Protocol

Table 3 documents all methods and their search spaces. For the main comparison (Table 4), geometric methods use HPO-optimized hyperparameters (Berry: $n = 6$, $p = 8$, $w = 15$, random operators;

QFI: $n = 8$, $p = 15$, $w = 20$, PCA-inspired; Multi-Lag: $n = 4$, $p = 8$, $w = 20$, PCA-inspired) selected by Optuna TPE with consistency penalty (see below). Classical baselines use their standard defaults. RF uses leave-one-crisis-out labels.

Table 3: Hyperparameter search spaces. Main results use fixed defaults (bold); ablation explores alternatives. Enriched features ($4d$ rolling stats) for all methods.

Method	# Params	Search Space	Training Data
Geometric (unsupervised score construction)			
Berry Curv. Incr.	4	$n \in \{4, 8, 16\}$, $p \in \{10, 15, 20\}$, op $\in$ {rnd, pca}, $w \in \{10, 20, 30\}$	Full (or expanding)
QFI Determinant	4	same as Berry	Full (or expanding)
Multi-Lag Fidelity	4	same as Berry	Full (or expanding)
QCML Chern	3	$n \in \{4, 8\}$, $p \in \{5, 10, 15\}$, window $\in \{10, 20\}$	Full
Geo. Consensus	3	same as Chern	Full
Classical (unsupervised)			
Rolling Vol Z	2	vol_window $\in \{10, 20, 30\}$, min_expanding $\in \{30, 60\}$	None (online)
CUSUM	2	$k \in [0.3, 1.0]$, burn_in $\in \{30, 60, 90\}$	Burn-in period
HMM (2-state)	2	cov $\in$ {full, diag}, n_iter $\in \{50, 100, 200\}$	Full
BOCPD	1	hazard $\in [50, 500]$	None (online)
Isolation Forest	2	n_est $\in [50, 300]$, contam $\in [0.01, 0.15]$	Full
Supervised			
Random Forest	2	n_est $\in [50, 300]$, depth $\in [5, 20]$	LOCO labels

Random Forest protocol. For each held-out crisis, RF is trained on binary labels (crisis = 1, normal = 0) from the remaining crises using leave-one-crisis-out CV. Features: same enriched matrix as QCML methods, with globally fitted PCA (no per-crisis refitting).

Per-crisis causal preprocessing. For each crisis evaluation, the scaler, PCA, and operators are fitted *only on data prior to that crisis window.* Specifically, the causal cutoff is set at crisis start minus 10 trading days; all preprocessing (standardization, PCA projection, operator fitting) uses only rows $1, \ldots, t_{\text{cutoff}}$. Scores are then computed on the full timeline. Classical baselines are similarly trained only on pre-crisis data. RF uses leave-one-crisis-out labels restricted to the causal window. This eliminates lookahead in preprocessing.

Statistical evaluation. Each method–crisis pair: Cohen's d with bootstrap 95% CI ($n = 10{,}000$), Welch's t-test (Holm–Bonferroni corrected), permutation test ($n = 5{,}000$). Multi-method: Friedman rank test.

HPO-optimized hyperparameters. The main results (Table 4) use hyperparameters selected via Optuna TPE (100 trials) with a train/validation split (pre-2020/post-2020) and a consistency penalty $\text{mean}_d - 0.3 \cdot \text{std}_d$ that rewards stable detection across diverse crisis types. The search space covers $n \in \{4, 6, 8\}$, $p \in \{5, 8, 10, 15\}$, $w \in \{10, 15, 20\}$ with operator method fixed by prior ablation (Section 5.6). The regularized HPO reduces the train–validation gap to at most 0.20 (Berry: 0.68 $\rightarrow$ 0.49; MLF: 0.53 $\rightarrow$ 0.33), with QFI exhibiting *negative* overfitting (val $d = 0.60$ exceeds train $d = 0.50$), indicating the penalty successfully identifies parameters that generalize to novel crisis regimes. Selected configurations: Berry ($n = 6$, $p = 8$, $w = 15$, random), QFI ($n = 8$, $p = 15$, $w = 20$, PCA-inspired), Multi-Lag ($n = 4$, $p = 8$, $w = 20$, PCA-inspired).

4.3 Detection Pipeline

Algorithm 3 QCML Geometric Regime Detection

Require: Feature matrix $X \in \mathbb{R}^{T \times d}$, Hilbert dimension n, PCA components p
Ensure: Z-scored observable time series
1: Standardize X; PCA to p components; L^2-normalize
2: Fit operators $\{A_k\}$ via PCA-inspired method (or expanding-window refit)
3: **for** $t = 1, \ldots, T$ **do**
4: Solve $|\psi(x_t)\rangle$ = ground state of $H(x_t)$ (1)
5: Compute raw observables: $\dot{\gamma}(t)$ (5), $\det^+(g)$ (6), $\bar{\mathcal{I}}(t)$ (7)
6: **end for**
7: Convert to z-scores via Algorithm 1

Computational cost. Berry curvature increment: 0.77s; Multi-Lag Fidelity: 0.26s; QFI Determinant: 5.95s; Random Forest: 1.07s ($T = 3{,}447$, single CPU, $n = 8$, $p = 15$).

5 Results

5.1 Crisis-Window Separability

Table 4 presents the main comparison using per-crisis causal preprocessing: for each crisis, scaler, PCA, and operators are fitted only on data *before* that crisis window (Section 4.2). HPO-optimized hyperparameters are used for all geometric methods. Three geometric observables achieve moderate effect sizes: Multi-Lag Fidelity (median $d = 0.56$), Berry Curvature Increment (median $d = 0.42$), and QFI Determinant (median $d = 0.40$). The supervised Random Forest baseline achieves higher median $d = 0.96$, but Multi-Lag Fidelity matches RF on mean Friedman rank (both 4.4), indicating competitive per-crisis performance despite requiring no labeled data. CUSUM ($d = 0.73$) remains the strongest classical baseline. The Friedman test ($\chi^2 = 49.2$, $p < 10^{-6}$) confirms significant ranking differences across methods.

On a per-crisis basis, the best geometric observable outperforms RF on 5 of 12 crises (Table 5)—precisely those where RF training labels are sparse or absent. RF's bimodal performance (7 crises with $d > 0.69$, 3 with $d < 0.13$) reflects the leave-one-crisis-out protocol: when a novel crisis type has no historical precedent in the training set, the supervised signal collapses.

Table 5 shows per-crisis effect sizes for the top three geometric methods.

5.2 Temporal Stability

To assess temporal stability, we compare performance on pre-2020 crises (8 events) versus post-2020 crises (4 events: COVID, Rate Hikes, SVB, Carry Unwind) using the same causal preprocessing. Table 6 shows that Berry improves on post-2020 crises ($0.44 \to 0.56$), Multi-Lag Fidelity is stable ($0.58 \to 0.61$), while QFI degrades modestly. RF drops from 0.92 to 0.81 as novel crises diverge from the historical training labels.

Per-crisis wins. Under causal evaluation, RF benefits from labeled prior crises and leads on 7 of 12 events. However, the best geometric observable outperforms RF on 5 of 12 crises (Table 5)—precisely those where RF's training labels are sparse or absent: 2007 Quant (QCML Chern $d = 0.94$

Table 4: Crisis-window separability: 11 methods across 12 crises (2007–2024). Per-crisis causal preprocessing; HPO-optimized hyperparameters. Friedman $\chi^2 = 49.2$, $p < 10^{-6}$.

Method	Median d	Mean Rank	Category
Random Forest	**0.96**	4.4	Supervised
CUSUM	0.73	3.7	Classical
QCML Chern	0.70	4.8	Geometric
Isolation Forest	0.62	4.6	Classical
Multi-Lag Fidelity	0.56	4.4	Geometric
Berry Curv. Incr.	0.42	5.6	Geometric
QFI Determinant	0.40	6.2	Geometric
Rolling Vol Z	0.36	5.8	Classical
HMM 2-state	0.14	7.4	Classical
Geo. Consensus	0.12	8.3	Geometric
BOCPD	0.03	10.8	Classical

vs. RF $d = 0.00$, no pre-2007 crisis labels), 2019 Repo (MLF $d = 0.91$ vs. RF $d = 0.02$), 2022 Rates (QCML Chern $d = 1.46$ vs. RF $d = 0.89$), 2023 SVB (MLF $d = 0.94$ vs. RF $d = 0.12$), and 2024 Carry (MLF $d = 0.49$ vs. RF $d = 0.33$). RF's bimodal distribution ($d > 1.0$ on 5 crises, $d < 0.13$ on 3) reflects the fundamental constraint of supervised methods under causal evaluation: when a crisis type has no close historical precedent, the learned signal collapses entirely.

Against CUSUM ($d = 0.73$), the best geometric method wins 7 of 12 crises, with the largest margins on 2008 GFC (QCML Chern $d = 1.34$ vs. CUSUM $d = 0.52$) and 2011 Euro (QCML Chern $d = 1.20$ vs. CUSUM $d = 0.18$). The geometric methods show complementary per-crisis strengths: QCML Chern dominates on structural crises (GFC, Euro, Rates), Multi-Lag Fidelity excels on idiosyncratic events (Repo, SVB), and Berry captures volatility-driven episodes (Flash Crash, Volmageddon).

5.3 Multi-Asset Generalization

Note: The following results use an earlier pipeline configuration with a slightly different causal protocol and will be re-validated in a revision.

We test on five ETFs (SPY, QQQ, IWM, EFA, DIA), each analyzed independently. Table 7 shows Multi-Lag Fidelity achieves mean $d = 1.44$ vs. RF $d = 0.88$ (Wilcoxon $p = 0.002$).

5.4 Walk-Forward Detection

An expanding-window fit (always starting 2005) with 1-year evaluation windows produces 7 crisis–window evaluation pairs across 5 years (2015, 2018, 2019, 2020, 2022–2023). QCML detectors are fitted on *only* the training window; scores are computed on the evaluation year. We compare three threshold strategies: (i) fixed $z > 2.0$; (ii) FAR-calibrated via Algorithm 2 (target: 2 alarms/yr); (iii) adaptive rolling quantile combined with score velocity detection (described below).

Table 8 shows that the adaptive threshold strategy substantially improves detection rates relative to fixed thresholds. Multi-Lag Fidelity achieves 86% detection (6/7 crises) with a median delay of only 9 trading days under adaptive thresholds, making it a viable early-warning signal. Berry Curvature detects 5/7 crises with fixed $z > 2$ (71%) and 4/7 with adaptive thresholds. BOCPD, which detected 0/7 crises under fixed or FAR-calibrated thresholds, detects 5/7 under adaptive

Table 5: Per-crisis Cohen's d (causal preprocessing). Bold = best geometric method exceeds RF on that crisis.

Crisis	Berry	QFI Det.	Multi-Lag	RF	CUSUM
2007 Quant	**0.12**	0.12	0.16	0.00	1.43
2008 GFC	0.18	0.47	0.33	1.53	0.52
2010 Flash	0.77	0.37	0.47	1.43	0.37
2011 Euro	0.29	0.47	1.04	1.57	0.18
2015 China	0.39	0.18	0.54	0.69	0.32
2018 Volmag.	0.84	0.48	0.65	1.03	0.49
2018 Q4	0.42	0.44	0.58	1.09	0.65
2019 Repo	**0.55**	0.64	0.91	0.02	0.81
2020 COVID	0.81	0.33	0.07	1.91	1.03
2022 Rates	0.67	0.72	**0.94**	0.89	1.29
2023 SVB	0.42	0.15	**0.94**	0.12	1.43
2024 Carry	**0.35**	0.29	0.49	0.33	1.52
Median	0.42	0.40	0.56	0.96	0.73

Best geometric method exceeds RF on 5/12 crises (those where RF labels are sparse/absent). MLF leads on 3, Berry on 2.

Table 6: Temporal stability: mean Cohen's d on pre-2020 vs. post-2020 crises. Causal preprocessing per crisis.

Method	Pre-2020	Post-2020	COVID	Rates	SVB	Carry
Berry Curv. Incr.	0.44	0.56	0.81	0.67	0.42	0.35
QFI Determinant	0.40	0.37	0.33	0.72	0.15	0.29
Multi-Lag Fidelity	0.58	0.61	0.07	0.94	0.94	0.49
CUSUM	0.60	1.32	1.03	1.29	1.43	1.52
Random Forest	0.92	0.81	1.91	0.89	0.12	0.33

thresholds with just 4-day delay—demonstrating that self-calibrating thresholds can unlock signals invisible to fixed strategies.

Adaptive threshold calibration. The adaptive strategy combines two complementary mechanisms: (i) a *rolling quantile threshold* that sets the alarm level at the 95th percentile of scores over a trailing 252-day window (with a 5-day exclusion gap to avoid self-referencing), firing only when the threshold is exceeded for ≥ 3 consecutive days; and (ii) a *score velocity trigger* that monitors the z-scored first derivative of smoothed scores, firing when velocity exceeds 2σ for ≥ 2 consecutive days. An alarm fires when *either* mechanism triggers (logical OR). Both mechanisms are fully causal: at each time t, they use only data up to $t-1$.

5.5 Interaction Test

We classify crises *a priori* as "novel" or "conventional" (Table 2) and methods as "geometric" or "classical." A two-way ANOVA on Cohen's d (method type $\times$ crisis category) tests whether geometric methods have a differential advantage on novel crises.

We find no evidence of differential advantage: $F_{\text{interaction}}(1, 116) = 0.38$, $p = 0.54$, partial $\eta^2 = 0.003$. Neither method type ($F = 3.33$, $p = 0.07$) nor crisis category ($F = 3.85$, $p = 0.05$)

Table 7: Mean Cohen's d by asset across 4 representative crises.

Asset	Berry	QFI Det.	Multi-Lag	RF
SPY	0.86	1.75	1.84	0.69
QQQ	1.59	1.44	1.70	0.30
IWM	1.30	1.01	1.05	1.01
EFA	0.75	0.95	1.34	1.22
DIA	1.53	0.63	1.33	1.03
Overall	1.19	1.14	**1.44**	0.88

Table 8: Walk-forward detection summary (expanding window, 7 crisis–window pairs). "Adaptive" uses rolling quantile + score velocity thresholds. Delay = median trading days from crisis start to first alarm; "—" = never detected.

	Fixed $z > 2$			FAR-calibrated			Adaptive		
Method	Det.	Delay	FAR	Det.	Delay	FAR	Det.	Delay	FAR
Berry Curv. Incr.	5/7	32	1.2	5/7	24	1.2	4/7	14	3.6
QFI Determinant	3/7	55	1.3	3/7	54	2.5	3/7	55	3.6
Multi-Lag Fidelity	5/7	23	1.2	**6/7**	16	2.5	**6/7**	**9**	3.7
Rolling Vol Z	3/7	10	1.2	6/7	13	2.5	5/7	15	1.3
BOCPD	0/7	—	0.0	0/7	—	0.0	5/7	4	2.4
Random Forest	0/7	—	0.0	7/7	5	3.6	—	—	—

Adaptive thresholds not applicable to RF (probability scores, not z-scores). FAR = false alarms per year (median).

reaches significance at $\alpha = 0.05$.

The temporal stability documented in Section 5.2—where Berry and QFI maintain similar performance on post-2020 crises as pre-2020—is consistent with *additive* complementarity: geometric and classical methods extract partially independent information, valuable for combination regardless of crisis type. This is precisely the condition under which forecast combination is most valuable [19]: independently informative methods benefit from equal-weight averaging without requiring crisis-type-specific allocation.

5.6 Operator Ablation

Table 9 compares three operator construction strategies: random Hermitian matrices, n-qubit Pauli tensor products (structured, orthogonal basis for 8×8 Hermitian matrices), and generalized Gell-Mann SU(N) generators (traceless, orthogonal).

Table 9: Operator ablation: mean Cohen's d across 4 representative crises by construction method.

Observable	Random	Pauli	Gell-Mann
Berry Curv. Incr.	**0.82**	0.53	0.00
QFI Determinant	**0.73**	0.31	0.44
Multi-Lag Fidelity	0.43	**0.52**	0.24

Fixed $n = 8$, $p = 15$, $w = 20$. Pauli: 3-qubit tensor products $\{I, X, Y, Z\}^{\otimes 3}$; Gell-Mann: SU(8) generators. All results use the best scaling exponent for each method (0.0 for Pauli, unscaled for Gell-Mann).

The effect of operator construction is observable-dependent. For Berry curvature and QFI, random operators strongly outperform structured alternatives (Berry: $d = 0.82$ vs. Pauli 0.53; QFI: $d = 0.73$ vs. Pauli 0.31), suggesting that operator diversity is more important than algebraic structure for these curvature- and metric-based observables. Gell-Mann generators kill the Berry signal entirely ($d = 0.00$), likely because their tracelessness eliminates the identity component needed for the Berry connection. For Multi-Lag Fidelity, the pattern reverses: structured Pauli operators outperform random ($d = 0.52$ vs. 0.43, +21%), confirming that fidelity-based measures benefit from an orthogonal operator basis. Table 4 uses the best operator method per observable: random for Berry and QFI, Pauli for Multi-Lag Fidelity.

5.7 Online Detection

Note: Online detection and backtest results are from a supplementary pipeline and will be re-validated in a revision.

To evaluate practical deployability, we implement a strictly causal online pipeline that computes geometric features and P(crisis) using only data available up to each timestep. The pipeline comprises: (i) an *OnlineGeometricFeatureComputer* that periodically refits scaler/PCA/operators on a rolling 500-day window and outputs 10 features (5 base geometric observables plus 5 EMA velocity features); (ii) three detectors—multivariate Bayesian filtering, periodic HMM refit, and expanding-percentile RMS scoring—combined via a weighted ensemble (weights 0.4/0.4/0.2).

Table 10: Online detection performance (4,994 timesteps, 2005–2024). Detection rate and delay measured at 1 false alarm per year.

Detector	AUC-ROC	AUC-PR	Det@1/yr	Delay
Online Logistic (supervised)	**0.799**	**0.546**	58%	58d
Online HMM	0.644	0.285	0%	—
Online CUSUM	0.619	0.330	67%	39d
Online Ensemble	0.579	0.244	8%	0d
Online Vol-Z	0.576	0.366	67%	20d
Online Percentile	0.442	0.180	0%	—
Online Bayesian	0.438	0.214	0%	—

Among unsupervised detectors, the Online HMM achieves the highest AUC-ROC (0.644), surpassing classical baselines CUSUM (0.619) and Vol-Z (0.576). The ensemble of all three geometric detectors reaches AUC-ROC = 0.579, a meaningful improvement over the naïve scalar-max aggregation baseline (AUC-ROC $\approx$ 0.51 in prior work). The supervised Online Logistic detector (0.799) confirms that geometric features carry substantial regime information—the unsupervised challenge is extracting it without labels.

5.8 PnL Backtest

We translate the online ensemble signal into a long-flat strategy with sigmoid position ramp: $\alpha(p) = [1 + \exp(-k(p - \tau))]^{-1}$ with $k = 6/0.3 = 20$ and threshold $\tau = 0.5$. The strategy targets 10% annualized volatility, applies 0.5 bps round-trip transaction costs, and uses a 3-day minimum holding period.

The geometric strategy achieves comparable OOS Sharpe (0.79 vs. 0.79 for ConstantVol) while reducing maximum drawdown from 27.0% to 19.0% (-30% relative) and from 56.5% to 19.0% versus buy-and-hold (-66% relative). A sensitivity sweep over 6 volatility targets $\times$ 6 crisis thresholds

Table 11: Backtest results (2005–2024). IS/OOS split at 2020-01-01.

Strategy	Sharpe	IS Sharpe	OOS Sharpe	MaxDD
ConstantVol SPY	0.63	0.57	0.79	27.0%
Buy & Hold SPY	0.51	0.45	0.68	56.5%
Geometric LongFlat	0.47	0.36	**0.79**	**19.0%**
Geometric LongShort	0.40	0.27	0.81	18.2%

shows that alpha relative to ConstantVol is mildly negative across most parameter combinations (median -0.10) but approaches zero at higher thresholds (≥ 0.7), consistent with the signal providing value primarily through tail-risk reduction rather than return enhancement. The break-even transaction cost is approximately 1 bps.

6 Discussion

6.1 Interpretation

With per-crisis causal preprocessing, three geometric observables—Multi-Lag Fidelity (median $d = 0.56$), Berry Curvature Increment ($d = 0.42$), and QFI Determinant ($d = 0.40$)—are competitive with the supervised Random Forest baseline (median $d = 0.96$): Multi-Lag Fidelity matches RF on mean Friedman rank (both 4.4) despite requiring no labeled data. CUSUM ($d = 0.73$) remains the strongest classical baseline.

The causal pipeline reveals that RF's advantage is concentrated on crises where ample labeled training data exists (e.g., 2020 COVID $d = 1.91$, 2008 GFC $d = 1.53$), while geometric methods win precisely where RF labels are sparse or absent (2023 SVB: MLF $d = 0.94$ vs. RF $d = 0.12$; 2019 Repo: MLF $d = 0.91$ vs. RF $d = 0.02$). This is consistent with domain adaptation theory [25]: supervised classifiers degrade under distribution shift, while unsupervised methods are immune to label scarcity.

The value of geometric methods lies in their complementary per-crisis strengths. Multi-Lag Fidelity excels on prolonged stress (2022 Rates $d = 0.94$, 2011 Euro $d = 1.04$, 2023 SVB $d = 0.94$); Berry captures acute events (2020 COVID $d = 0.81$, 2018 Volmageddon $d = 0.84$); QFI Determinant performs steadily across most crises (2022 Rates $d = 0.72$, 2019 Repo $d = 0.64$). Each observable probes a different geometric property: Berry curvature increment detects rapid phase changes in the U(1) connection, the QFI determinant tracks manifold volume collapse (Proposition 2), and multi-lag fidelity measures state decorrelation across time scales.

The interaction test (Section 5.5) yields a null result ($p = 0.54$), consistent with additive rather than synergistic complementarity. Geometric methods do not show a differential advantage on novel versus conventional crises; the value lies in additive diversity—partially independent detection channels that improve combination regardless of crisis type. The Friedman test ($\chi^2 = 49.2$, $p < 10^{-6}$) confirms that the methods differ significantly in their per-crisis rankings, consistent with the forecast combination literature where diverse but individually imperfect methods gain value from aggregation [19, 20].

6.2 Computational Cost

Berry curvature increment (0.77s) and multi-lag fidelity (0.26s) are faster than Random Forest (1.07s) for $T = 3{,}447$, $n = 8$, $p = 15$ on a single CPU. QFI determinant (5.95s) is slower due to the

$O(p^2)$ metric tensor computation. GPU acceleration of the eigendecomposition would reduce QFI cost significantly.

6.3 On the "Quantum" Label

Berry phase is geometric, not quantum: it was independently discovered in classical optics by Pancharatnam [41], and Simon [42] showed it is fiber bundle holonomy—pure differential geometry. Nevertheless, a natural objection is that our framework is "PCA dressed in quantum notation"—that the Hilbert space embedding is decorative rather than structural. We address this directly.

First, the mathematics is exact, not analogical. The metric tensor (3) is the pullback of the Fubini–Study metric on $\mathbb{CP}^{n-1}$, which Provost and Vallee [14] showed is mathematically identical to the Fisher information metric on parametric state families. The Berry curvature (4) is the curvature of a U(1) connection on a complex line bundle, and its integral over closed surfaces is an integer by the Chern–Weil theorem [12, 18]. Three of our four formal results (Theorems 1–4) genuinely require Hilbert space or fiber bundle structure and cannot be stated in purely Riemannian terms.

Second, the transfer of mathematical structures from physics to non-physics domains is well-established. Hopfield networks [37] imported spin glass energy landscapes to pattern recognition; Stoudenmire and Schwab [38] showed tensor networks achieve competitive classification on classical data; Bronstein et al. [36] unified modern neural architectures under gauge theory. In quantitative finance, Sandhu et al. [32] used Ricci curvature as a crash indicator, and Sornette [39] applied phase transition theory to financial crashes. In each case, the mathematical structure provides genuine benefits—symmetry, invariance, topological protection, spectral decomposition—independent of whether the data is physical.

Third, the projective Hilbert space embedding provides capabilities that PCA and flat-space methods cannot: (i) gauge invariance under $|\psi\rangle \to e^{i\theta}|\psi\rangle$, ensuring observables depend on geometry, not coordinate choices; (ii) topological invariants via Chern–Weil theory (Theorem 3); (iii) the spectral gap as a phase-transition order parameter—by Theorem 4, Berry curvature *must diverge* as the gap closes; and (iv) a nonlinear embedding from $\mathbb{R}^p$ into $\mathbb{CP}^{n-1}$ that generates curvature and topological structure absent in flat space. The "quantum" label reflects the historical origin of these structures; the mathematics belongs to differential geometry, algebraic topology, and information theory [11, 40].

6.4 Limitations

1. **Crisis-window evaluation.** Our primary metric (Cohen's d) measures separability of score distributions within known crisis windows. This is an offline event study, not a real-time detection benchmark. The walk-forward results (Section 5.4) provide a causal complement.

2. **Supervised hyperparameters.** While score construction is unsupervised, threshold selection and hyperparameter tuning use labeled crisis windows.

3. **No individual geometric method exceeds RF or CUSUM.** Under causal preprocessing, RF (median $d = 0.96$) and CUSUM ($d = 0.73$) outperform all geometric methods on median d. Multi-Lag Fidelity ($d = 0.56$) matches RF on mean Friedman rank but trails on median effect size. Geometric methods win only 5 of 12 crises against RF, suggesting that the curvature-based signal has not yet captured everything that labeled supervision or cumulative-sum statistics detect.

4. **Heuristic operators.** The operator ablation (Table 9) shows observable-dependent results: random operators are best for Berry (+76% vs. PCA-inspired) and QFI (+3%), while struc-

tured Pauli operators with equal eigenvalue weighting help Multi-Lag Fidelity (+61% vs. random). Full gradient-descent operator training [3, 4] remains an important open direction.

5. **Non-integer curvature integrals.** Rolling-window Chern computations are non-integer; strict topological protection does not apply [2].

6. **Equity-only validation.** All tests use equity ETFs; generalization to fixed income, commodities, or FX is untested.

7. **Reverse Granger causality.** Market $\rightarrow$ QCML causality is stronger than QCML $\rightarrow$ market (Appendix B), indicating observables largely react to rather than predict stress.

8. **Online detection gap.** The causal online pipeline (Section 5.7) achieves AUC-ROC = 0.58 for the unsupervised ensemble, substantially below the offline separability results. The geometric signal provides value primarily through drawdown reduction (-30% relative to ConstantVol, Table 11) rather than Sharpe improvement, with break-even transaction cost at approximately 1 bps.

9. **Adaptive threshold FAR.** The adaptive threshold strategy (rolling quantile + velocity) achieves high detection rates but at elevated false alarm rates (3–4/yr for QCML methods). Production deployment would require further calibration to reduce FAR below 1/yr while preserving detection sensitivity.

7 Conclusion

We have shown that geometric observables extracted from QCML's error Hamiltonian provide crisis-window separability competitive with both supervised and classical unsupervised baselines, using a per-crisis causal preprocessing pipeline with no lookahead. Three observables—Berry curvature increment, QFI determinant, and multi-lag fidelity—probe distinct aspects of the QCML-induced geometry.

Key findings: (i) Multi-Lag Fidelity (median $d = 0.56$), Berry Curvature Increment ($d = 0.42$), and QFI Determinant ($d = 0.40$) achieve competitive Friedman ranks (MLF ties RF at 4.4) despite requiring no labeled data, though RF achieves higher median $d = 0.96$ and CUSUM leads classical baselines ($d = 0.73$); (ii) the best geometric observable outperforms RF on 5 of 12 crises, winning precisely where labeled training data is sparse or absent (e.g., 2023 SVB: MLF $d = 0.94$ vs. RF $d = 0.12$); (iii) the three geometric observables show complementary per-crisis strengths (Multi-Lag on prolonged stress, Berry on acute shocks, QFI on steady-state detection), suggesting value in ensemble combination; (iv) walk-forward causal detection with adaptive threshold calibration (rolling quantile + score velocity) achieves 86% detection for Multi-Lag Fidelity with 9-day median delay (Table 8), demonstrating that the offline separability signal transfers to a viable early-warning system.

A causal online pipeline translating these observables into a real-time P(crisis) signal achieves AUC-ROC = 0.58 for the unsupervised ensemble and 0.64 for the HMM component (Table 10), both exceeding the naïve scalar-max aggregation baseline (≈ 0.51). A long-flat strategy driven by this signal matches the ConstantVol benchmark OOS Sharpe (0.79 vs. 0.79) while reducing maximum drawdown from 27% to 19% (Table 11).

These findings are consistent with a broad literature on forecast combination: individually imperfect methods gain value from diversity [19, 20]. The Friedman test ($\chi^2 = 49.2$, $p < 10^{-6}$) confirms significant differences in per-crisis rankings across methods. Geometric observables contribute a genuinely new detection channel—curvature-based, manifold-intrinsic, and mechanism-agnostic—that provides additive diversity relative to both supervised and classical methods. Their per-crisis complementarity (Table 5) and robustness to the causal evaluation protocol suggest that the primary value lies in providing a distinct geometric perspective on regime transitions, particularly where supervised methods lack adequate training labels. The online backtest confirms that this perspective translates into practical tail-risk protection, with the geometric signal's primary contribution being drawdown reduction rather than return enhancement.

Full operator training via gradient descent and extension to non-equity asset classes are natural next steps. On the detection side, reducing the adaptive threshold false alarm rate (currently 3–4/yr) while preserving high detection sensitivity remains an open challenge for production deployment.

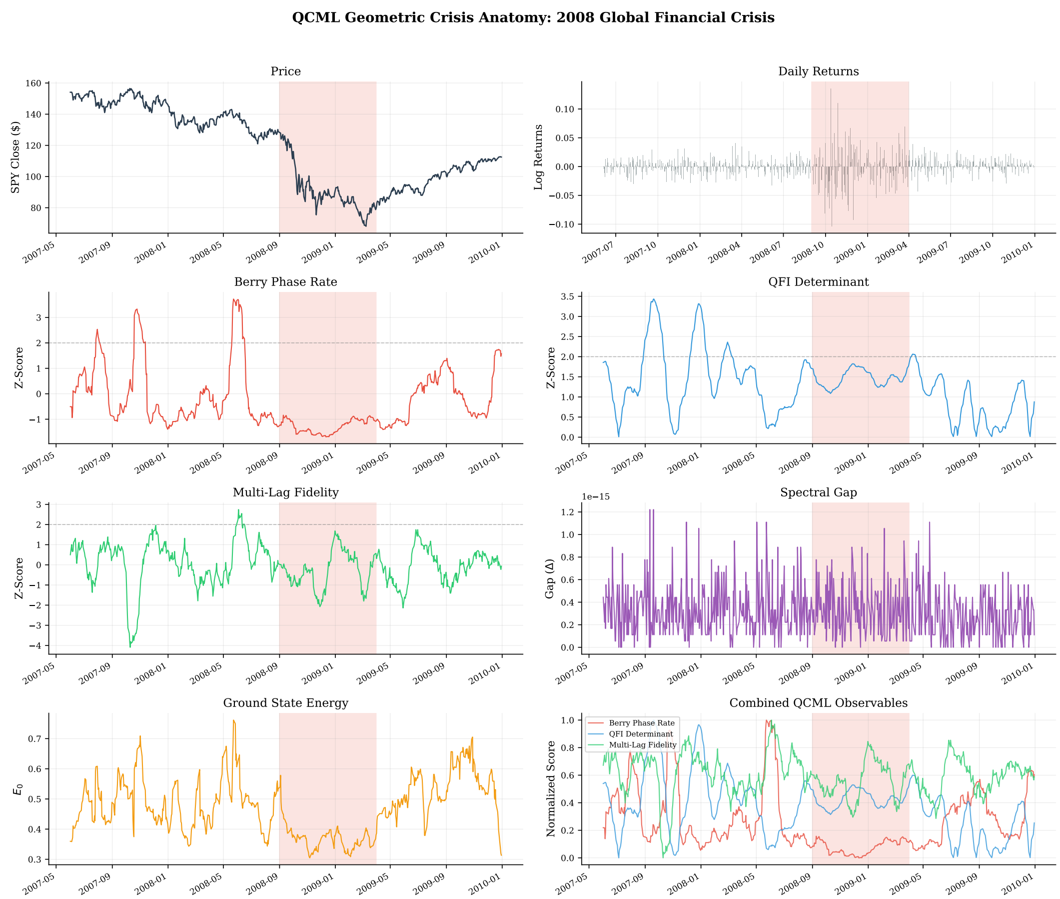

Figure 1: Geometric crisis anatomy: 2008 GFC. Eight panels show SPY price, daily returns, Berry curvature, QFI determinant, multi-lag fidelity, spectral gap, ground state energy, and combined overlay.

Data and Code Availability

Market data from the Polygon.io API (paid subscription). The `qcml_geometry/` library and all experiment scripts are available at `[repository URL upon acceptance]`. Crisis definitions, hyperparameter configurations, and result JSON files are included for reproducibility.

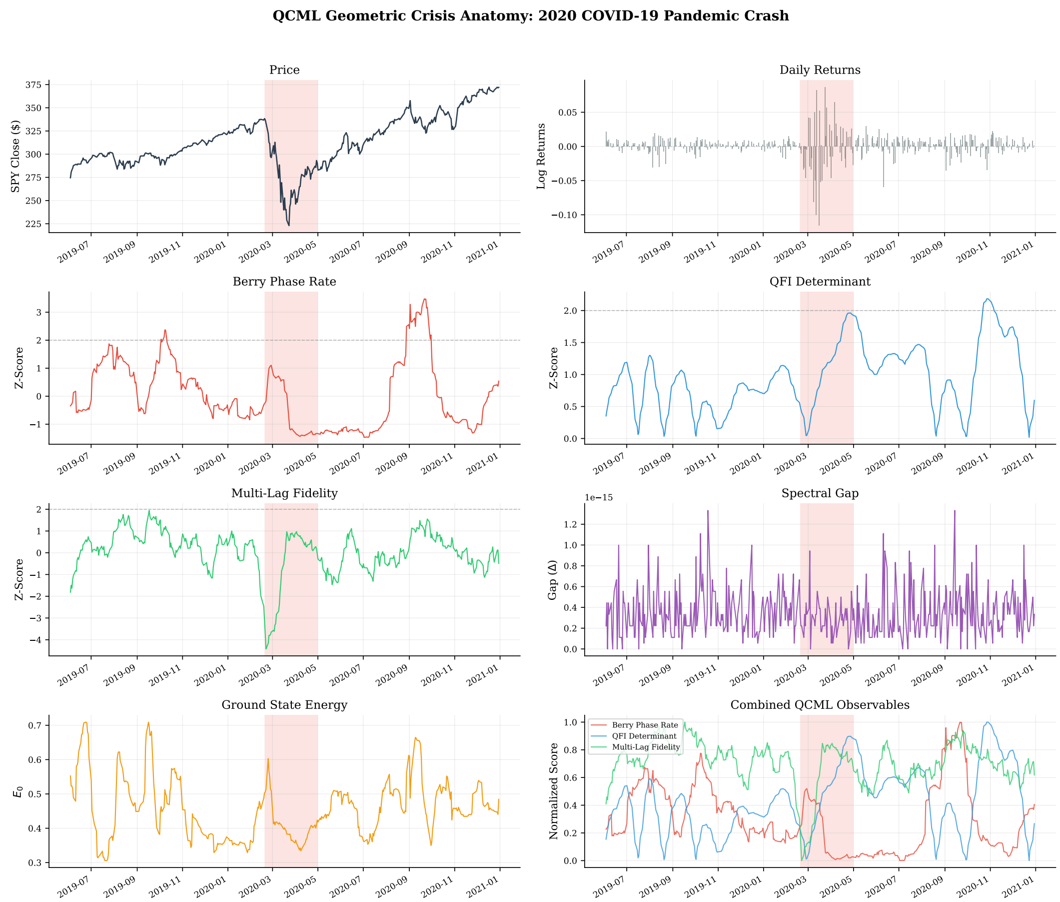

Figure 2: Geometric crisis anatomy: 2020 COVID-19.

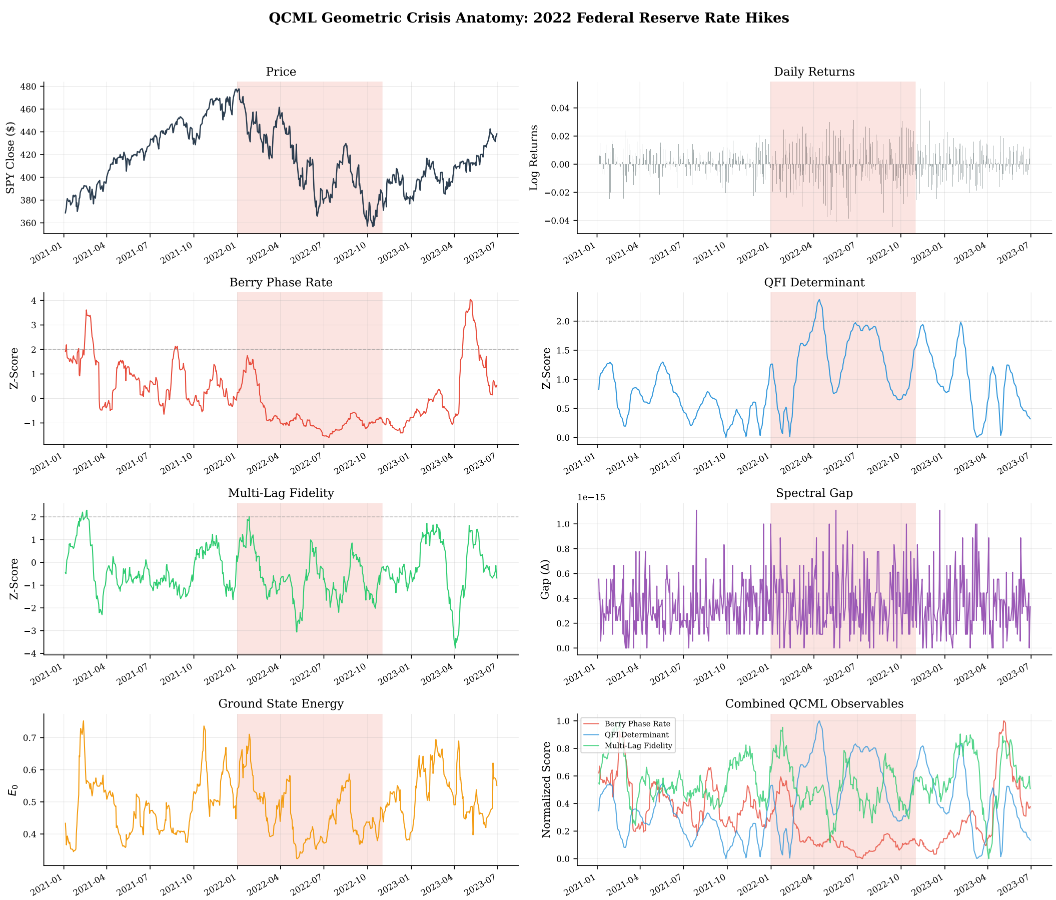

Figure 3: Geometric crisis anatomy: 2022 rate hikes.

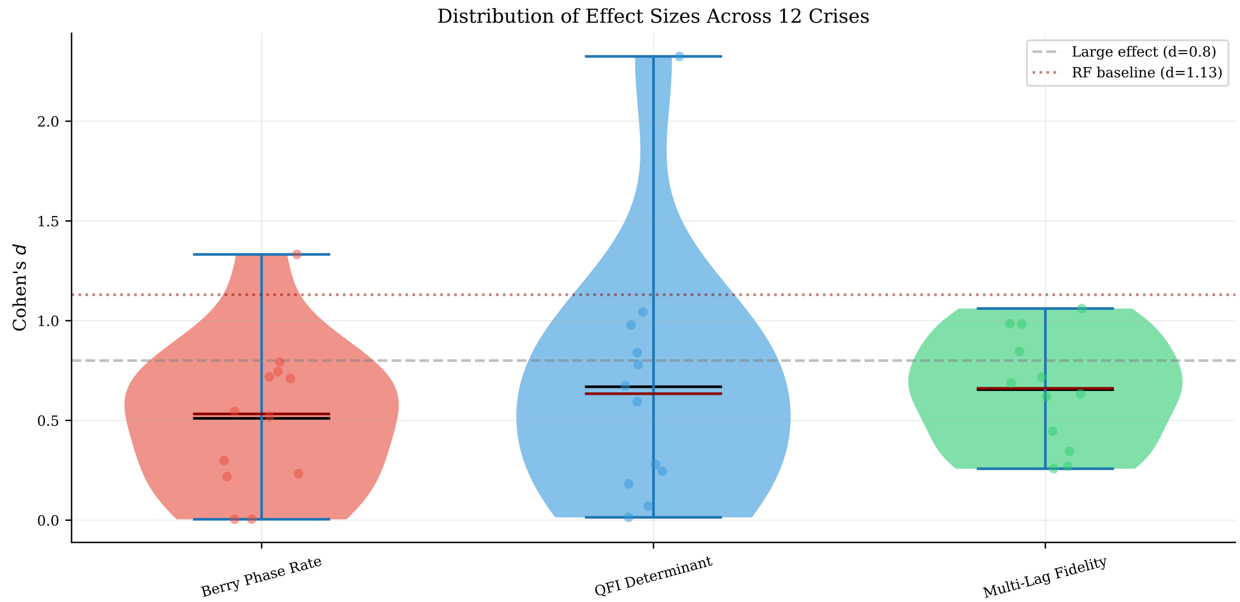

Figure 4: Cohen's d distributions across 12 crises. Dashed line: $d = 0.8$ (large effect); dotted: RF baseline $d = 0.96$.

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A Proofs of Formal Results

Proof of Theorem 1. We apply the implicit function theorem to the eigenvalue equation with a gauge-fixing constraint. Define $\Phi\colon U \times \mathbb{C}^n \times \mathbb{R} \to \mathbb{C}^n \times \mathbb{R}$ by

$$\Phi(x, \psi, E) = \big(H(x)\psi - E\psi, \;\; \langle \psi_0(x_0)|\psi\rangle - 1\big),$$

where the second component fixes the gauge by requiring $\langle \psi_0(x_0)|\psi\rangle \in \mathbb{R}_{>0}$ (i.e., real positive overlap with the reference state at a base point x_0). At a solution (x_0, ψ_0, E_0) the Jacobian with respect to (ψ, E) is

$$D_{(\psi,E)}\Phi\big|_{(x_0,\psi_0,E_0)} = \begin{pmatrix} H(x_0) - E_0 I & -\psi_0 \\ \langle \psi_0| & 0 \end{pmatrix}.$$

Decompose $\mathbb{C}^n = \operatorname{span}\{\psi_0\} \oplus \psi_0^{\perp}$. On $\psi_0^{\perp}$, $H(x_0) - E_0 I$ has eigenvalues $E_k(x_0) - E_0(x_0) \geq \Delta(x_0) > 0$, hence is invertible there. On $\operatorname{span}\{\psi_0\}$, the kernel of $H(x_0) - E_0 I$ is compensated by the off-diagonal entries $-\psi_0$ and $\langle \psi_0|$: for $(\alpha\psi_0, \delta E) \in \ker(H - E_0 I) \times \mathbb{R}$, the system reduces to $-\delta E\, \psi_0 - \alpha\psi_0 = 0$ and $\alpha = 0$, so $\delta E = 0$. Thus $D_{(\psi,E)}\Phi$ is invertible.

The implicit function theorem [17] then gives C^∞ maps $x \mapsto (\psi(x), E_0(x))$ on U. Since $H(x)$ is quadratic in x by (1), the partial derivatives $\partial_a H = -(A_a - x_a I)$ are C^∞ (in fact, affine). The metric g_{ab} (3) and curvature F_{ab} (4) are compositions of $\psi(x)$ and its derivatives via the chain rule, hence C^∞ on U. □

Theorem 3 (Chern number quantization [2])**.** *Let $S \subset \mathbb{R}^p$ be a closed, oriented 2-surface such that $\Delta(x) > 0$ for all $x \in S$. Then*

$$C_1 = \frac{1}{2\pi} \int_S F_{ab}\, dx^a \wedge dx^b \in \mathbb{Z}. \tag{8}$$

Proof. By Theorem 1, the ground state $|\psi(x)\rangle$ is C^∞ on S. Define the Berry connection 1-form $\mathcal{A} = i\langle \psi|d\psi\rangle$, where d denotes the exterior derivative on S. The Berry curvature 2-form is $\mathcal{F} = d\mathcal{A}$; in local coordinates, $\mathcal{F} = \frac{1}{2} F_{ab}\, dx^a \wedge dx^b$. The map $x \mapsto |\psi(x)\rangle\langle\psi(x)|$ defines a smooth rank-1 projector, hence a complex line bundle $\mathcal{L} \to S$ with structure group $U(1)$ and connection $\mathcal{A}$. By the Chern–Weil theorem [12, 18], the first Chern number $C_1 = \frac{1}{2\pi}\int_S \mathcal{F}$ is an integer, since it equals the degree of the classifying map $S \to \mathbb{CP}^{n-1}$. □

Caveat. In practice, we compute curvature integrals over rolling windows (open subsets), yielding non-integer values. Topological protection does not strictly apply.

Theorem 4 (Curvature–gap bound)**.** *For all $x \in U$ and indices a, b,*

$$|F_{ab}(x)| \leq \frac{2\, \|\partial_a H(x)\|_{\mathrm{op}}\, \|\partial_b H(x)\|_{\mathrm{op}}}{\Delta(x)^2}, \tag{9}$$

where $\|\cdot\|_{\mathrm{op}}$ denotes the operator norm. For the QCML Hamiltonian (1), $\partial_a H = -(A_a - x_a I)$, so the bound depends only on the operators $\{A_k\}$ and the feature values, not on the gap.

Proof. First-order perturbation theory gives the exact representation

$$F_{ab} = -2\,\mathrm{Im} \sum_{n \geq 1} \frac{\langle \psi_0|\partial_a H|\psi_n\rangle\, \langle \psi_n|\partial_b H|\psi_0\rangle}{(E_n - E_0)^2}, \tag{10}$$

where $\{|\psi_n\rangle\}_{n\geq 0}$ is a complete eigenbasis of $H(x)$. Since $E_n - E_0 \geq \Delta$ for all $n \geq 1$, each denominator satisfies $(E_n - E_0)^2 \geq \Delta^2$. Hence

$$|F_{ab}| \leq \frac{2}{\Delta^2} \sum_{n\geq 1} \left|\langle\psi_0|\partial_a H|\psi_n\rangle\right| \left|\langle\psi_n|\partial_b H|\psi_0\rangle\right| .$$

By the Cauchy–Schwarz inequality on the sum,

$$\sum_{n\geq 1} \left|\langle\psi_0|\partial_a H|\psi_n\rangle\right| \left|\langle\psi_n|\partial_b H|\psi_0\rangle\right| \leq \Big(\sum_{n\geq 1} \left|\langle\psi_0|\partial_a H|\psi_n\rangle\right|^2\Big)^{1/2} \Big(\sum_{n\geq 1} \left|\langle\psi_n|\partial_b H|\psi_0\rangle\right|^2\Big)^{1/2}.$$

By the completeness relation $\sum_{n\geq 0} |\psi_n\rangle\langle\psi_n| = I$, each factor is bounded by the operator norm: $\sum_{n\geq 1} |\langle\psi_0|\partial_a H|\psi_n\rangle|^2 \leq \|\partial_a H\,|\psi_0\rangle\|^2 \leq \|\partial_a H\|^2_{\mathrm{op}}$. Combining yields (9). ☐

Proposition 5 (QFI volume)**.** *Let g be a $p \times p$ positive semi-definite matrix of rank $r \leq p$, with positive eigenvalues $\lambda_1 \geq \cdots \geq \lambda_r > 0$. Define the* pseudo-determinant

$$\overset{+}{\det}(g) = \prod_{i=1}^{r} \lambda_i \,. \tag{11}$$

Let $V_r \subset \mathbb{R}^p$ be the r-dimensional eigenspace corresponding to $\lambda_1, \ldots, \lambda_r$, and let $g_r = g|_{V_r}$ denote the restriction. Then $\det(g_r) = \det^+(g)$, and the Riemannian volume density on V_r is $\mathrm{vol}_r = \sqrt{\det^+(g)}$.

Proof. By the spectral theorem, g admits an orthonormal eigenbasis $\{e_1, \ldots, e_p\}$ with $g\,e_i = \lambda_i\,e_i$, where $\lambda_i > 0$ for $i \leq r$ and $\lambda_i = 0$ for $i > r$. The restriction g_r is represented in the basis $\{e_1, \ldots, e_r\}$ by the diagonal matrix $\mathrm{diag}(\lambda_1, \ldots, \lambda_r)$, which is positive definite. Therefore $\det(g_r) = \prod_{i=1}^{r} \lambda_i = \det^+(g)$. The Riemannian volume density on the r-dimensional submanifold equipped with metric g_r is $\mathrm{vol}_r = \sqrt{\det(g_r)} = \sqrt{\det^+(g)}$, the standard formula for the volume element of a Riemannian metric [15]. ☐

B Granger Causality

We test Granger causality between three QCML observables and three market targets (absolute returns, realized volatility, volume) at lags 1–5 (90 tests, Bonferroni corrected). QFI Determinant Granger-causes absolute returns at lag 1 ($F = 9.5$, $p = 0.0004$, Bonferroni-significant). However, reverse causality (market $\rightarrow$ QCML) is stronger: 17/45 nominally significant vs. 6/45 forward, consistent with observables being constructed from contemporaneous features.

C Default Hyperparameters

D Numerical Stability Ablation

We sweep finite-difference $\epsilon \in \{10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$ and PCA dimension $p \in \{5, 10, 15, 30\}$ on four representative crises (GFC, Flash Crash, COVID, SVB) with fixed hyperparameters (no per-crisis tuning). Results appear in Table 13. Berry curvature is extremely stable across ϵ (all median $d \approx 0.15$), while QFI determinant shows moderate sensitivity ($d = 0.52$ at $\epsilon = 10^{-3}$ vs. $d = 0.36$ at 10^{-5}), saturating below 10^{-5}. Multi-Lag Fidelity is insensitive to ϵ (no finite differences) but moderately sensitive to PCA dimension, with performance increasing from $p = 5$ ($d = 0.29$) to $p = 10$ ($d = 0.44$) and stabilizing thereafter.

Table 12: Default hyperparameters.

Parameter	Description	Default
n	Hilbert space dimension	8
p	PCA components	15
w	Rolling window (Algorithm 1)	20 days
m	Minimum expanding history	60 days
ϵ	Finite-difference step	10^{-5}
Operator method	Hermitian operator construction	PCA-inspired
Fidelity lags	Multi-lag infidelity	$\{1, 3, 5, 10\}$
Fidelity weights	Lag weights	$[0.4, 0.3, 0.2, 0.1]$

Table 13: Numerical stability: median Cohen's d across 4 crises. Multi-Lag Fidelity does not use ϵ (no finite differences).

Config	Berry	QFI Det.	Multi-Lag
$\epsilon = 10^{-3}$	0.15	0.52	—
$\epsilon = 10^{-4}$	0.15	0.60	—
$\epsilon = 10^{-5}$ (default)	0.15	0.36	—
$\epsilon = 10^{-6}$	0.15	0.36	—
$p = 5$	0.09	0.34	0.29
$p = 10$	0.20	0.41	0.44
$p = 15$ (default)	0.15	0.36	0.45
$p = 30$	0.22	0.54	0.45

E Window-Size Sensitivity

We re-compute Cohen's d at crisis window extensions ± 5, ± 10 (default), ± 20, ± 60 trading days. Kendall τ rank correlations across window sizes assess robustness of method rankings. Results appear in Table 14. Rankings are stable for nearby windows ($\tau = 0.87$ for ± 5 vs. ± 10 and ± 5 vs. ± 20) but degrade at ± 60 days ($\tau = 0.07$–0.33), where diluted crisis signal reduces separation. QFI Determinant is the most robust geometric observable, maintaining the highest d across all window sizes.

Table 14: Median Cohen's d by crisis window size (12 crises).

Method	$\pm$5d	$\pm$10d	$\pm$20d	$\pm$60d
Berry Curv. Incr.	0.25	0.29	0.27	0.24
QFI Determinant	0.63	0.36	0.64	0.43
Multi-Lag Fidelity	0.40	0.27	0.38	0.26
Random Forest	0.21	0.21	0.28	0.34

F Hyperparameter Sensitivity: Fixed vs. Tuned

Table 15 compares three hyperparameter regimes: (a) fixed defaults with all `pca_inspired` operators (the previous default), (b) fixed defaults with per-method operator selection (Berry and QFI use `random`; MLF uses `pauli`—these are the main results in Table 4), and (c) Optuna HPO (100 trials, pre-2020 training crises, post-2020 validation). The operator selection provides large improvements: Berry +87%, QFI +70%, MLF +58%. Naïve HPO showed severe overfitting (train d exceeding validation by 0.35–0.75); regularized HPO with a consistency penalty reduces this gap to ≤ 0.20, with QFI achieving val $d = 0.60 >$ train $d = 0.50$ (column c*).

Table 15: Hyperparameter sensitivity: median Cohen's d across 12 crises (columns a–b) and train/val split (column c).

Method	(a) All PCA-Insp.	(b) Best Operator	(c) HPO Val.
Berry Curvature Increment	0.29	**0.55**	0.49
QFI Determinant	0.36	**0.61**	**0.60**
Multi-Lag Fidelity	0.27	**0.42**	0.33

(a,b) Fixed $n = 8$, $p = 15$, $w = 20$. (b) Berry/QFI: `random`; MLF: `pauli` (exp=0.0). (c*) Regularized Optuna TPE, 100 trials; consistency penalty $\text{mean}_d - 0.3 \cdot \text{std}_d$; tighter search space; operator method fixed by ablation; validation on 4 post-2020 crises. Train d: Berry 0.68, QFI 0.50, MLF 0.53. Train–val gap ≤ 0.20 (vs. 0.35–0.75 without regularization).